

# **BA1200 — Individual Coursework**

**Exploring Global Inequalities in Life Expectancy**

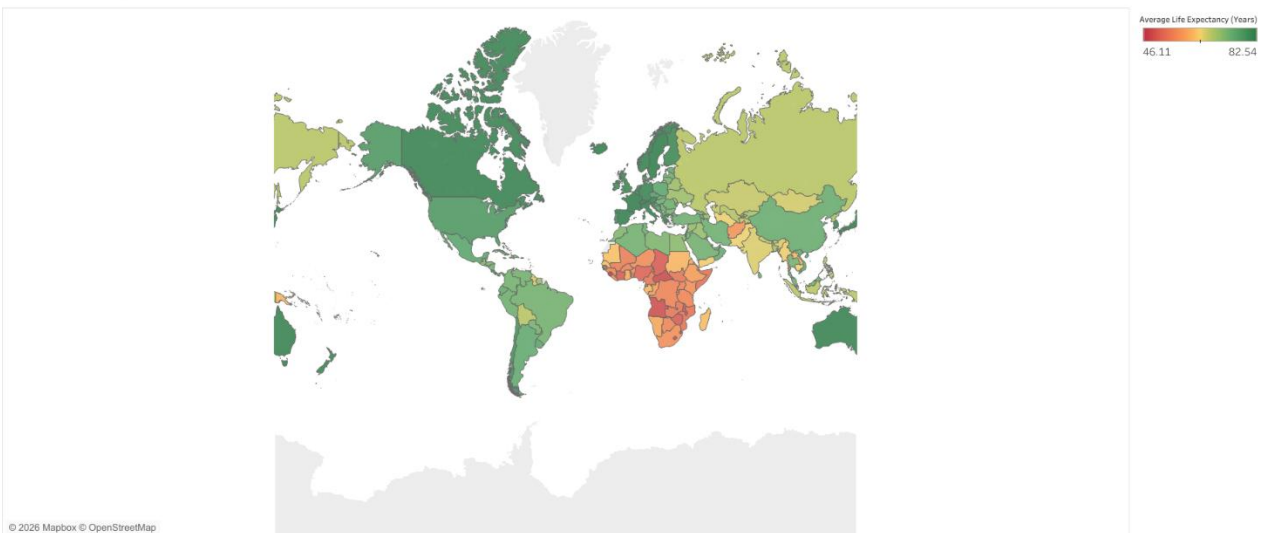
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## Visualization #1

Q1. Where are countries with the lowest life expectancies located and are they geographically clustered?

Life Expectancy Is Unevenly Distributed, with Lowest Levels Concentrated in Sub-Saharan Africa



What is the main insight of the chart? (MAX 200 WORDS) Why do you think this is insightful?

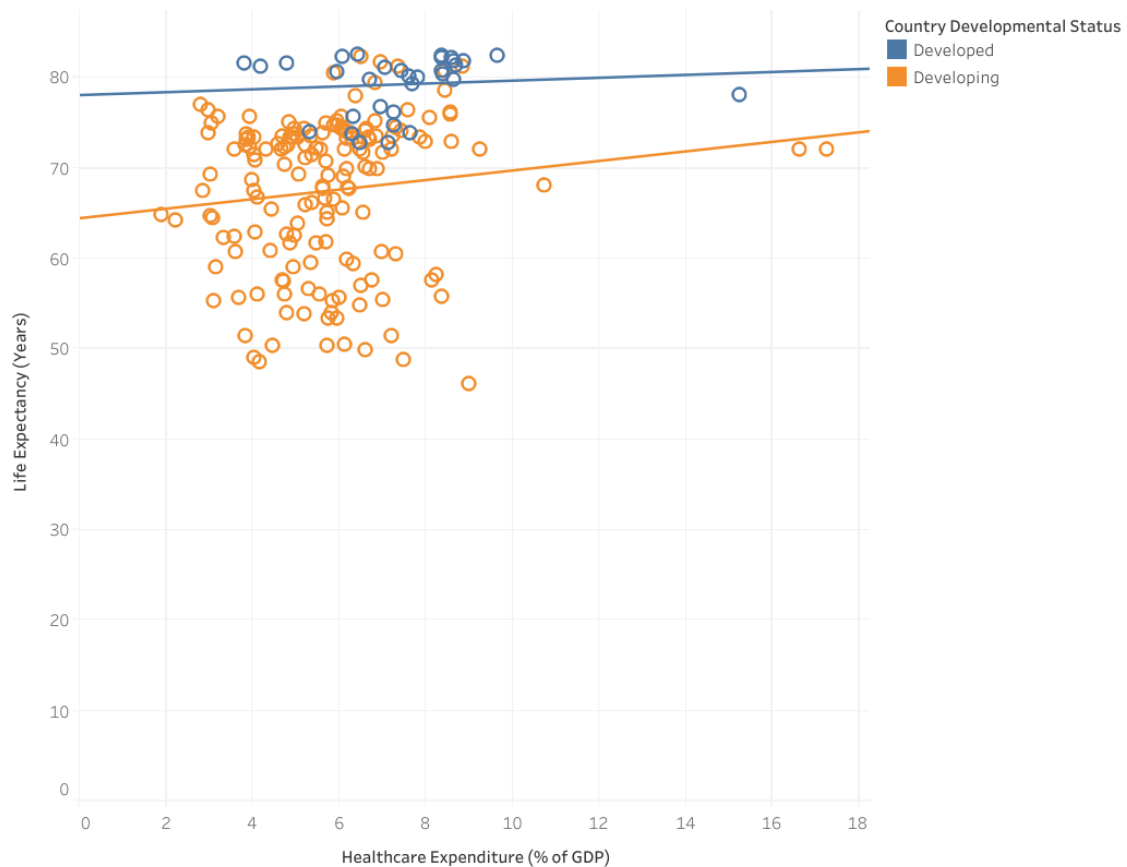
The choropleth map reveals clear global inequalities in life expectancy. A distinct geographical pattern is displayed, in which countries in Sub-Saharan Africa exhibit significantly lower life expectancies compared to those in Europe, North America and parts of East Asia such as Japan and South Korea. This clustering suggests that life expectancy is not randomly distributed but is closely associated with regional development. Lower life expectancy in Sub-Saharan Africa reflects the impact of limited healthcare infrastructure, weaker economic conditions and higher prevalence of infectious diseases. Furthermore, higher life expectancy in more developed regions reflects the benefits of sustained investment in healthcare systems, infrastructure, and social services.

This visualisation helps to build a foundation for further analysis of any socio-economic factors that influence life expectancy. However, while the map effectively identifies where inequality exists, it does not explain causation. Additionally, this visualisation does not account for within country variation or how life expectancies changed over time, which may limit the depth of the insight provided. The finding can however be used by international organisations to target investment in regions such as Sub-Saharan Africa, focusing on strengthening healthcare infrastructure.

## Visualization #2

Q2. To what extent does healthcare expenditure explain differences in life expectancy between countries?

Higher Healthcare Spending is Associated with Increased Life Expectancy but Shows Diminishing Returns



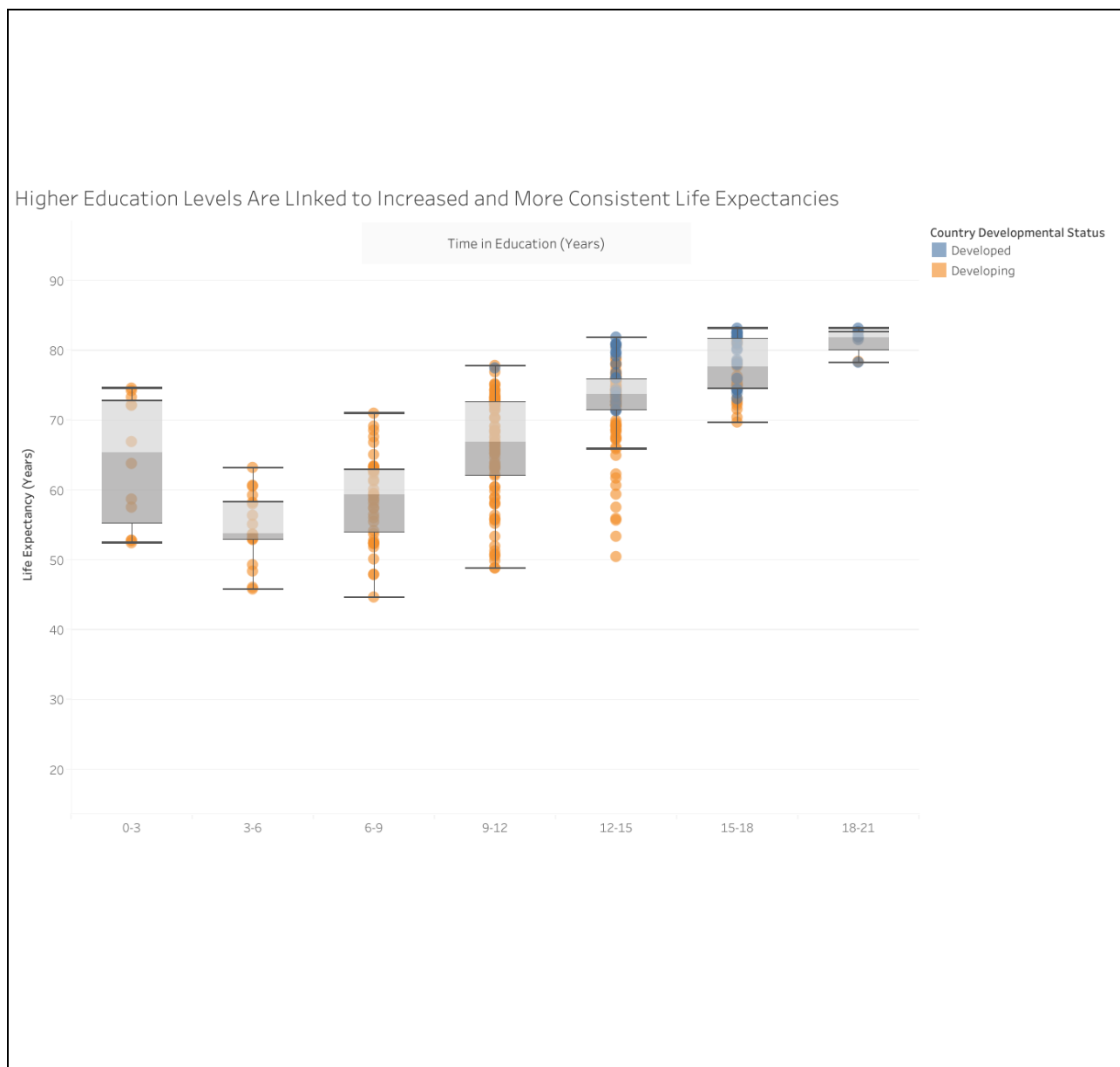
What is the main insight of the chart? (MAX 200 WORDS) Why do you think this is insightful?

This scatter plot explores the extent to which healthcare expenditure explains life expectancy inequalities. The visualisation reveals a positive relationship between healthcare expenditure (as a percentage of GDP) and life expectancy. This indicates that investment in healthcare is generally associated with improved population health. However, there is a clear distinction between developed and developing countries. Developed countries achieve higher life expectancies across similar ranges of expenditure, indicating an effectiveness of healthcare systems. In contrast, developing countries show greater variation. This suggests that healthcare spending alone is not sufficient to improve life expectancy. Furthermore, in developed countries, an increase in spending led to only a small improvement in life expectancy, implying limited returns at higher levels of expenditure.

This finding demonstrates that healthcare expenditure contributes to explaining global inequalities, but it is not sufficient on its own. Expenditure may not lead to improved outcomes due to issues such as misallocation of resources or limited access to care. Therefore, it is shown that not only the amount spent on healthcare matters, but also how effectively the money is used. This is useful for policymakers who should focus on not only increasing healthcare budgets but also on improving the efficiency of healthcare systems.

## Visualization #3

Q3. Does education help explain why some countries have lower life expectancy?



What is the main insight of the chart? (MAX 200 WORDS) Why do you think this is insightful?

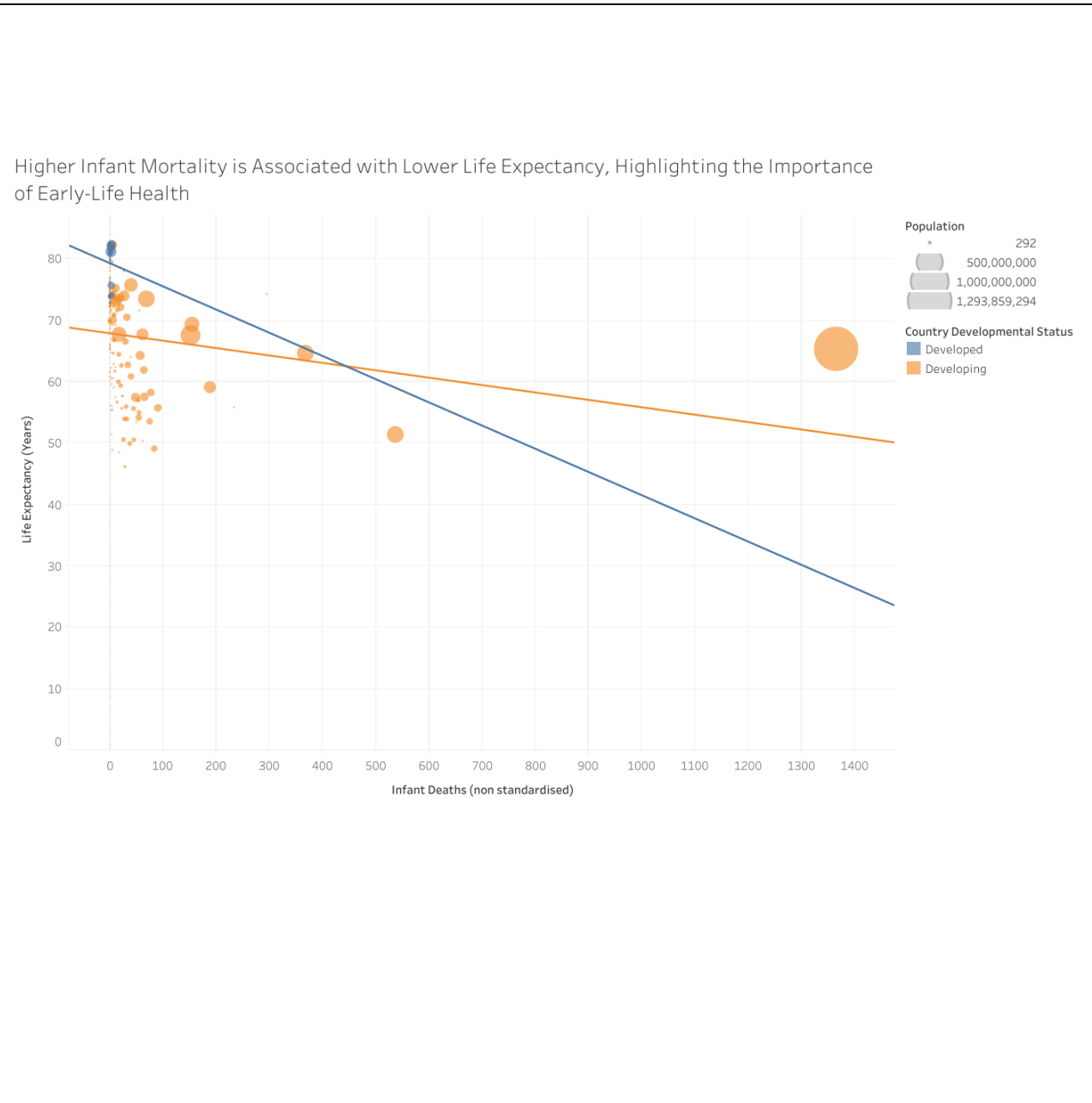
The boxplots demonstrate a clear positive relationship between years of schooling and life expectancy, with the median life expectancy generally increasing across higher education groups. However, it is also seen that the median is slightly higher in the 0-3 years group compared to the 3-6 and 6-9 groups. This difference may reflect the presence of a small number of outlier countries with relatively high life expectancies despite low formal education, possibly due to other factors such as strong healthcare systems, cultural practices or data limitations. In contrast, countries with higher levels of schooling (12+ years) show both higher and more consistent life expectancies. Furthermore, developed countries are concentrated at higher schooling categories and consistently achieve higher life expectancies within each group, while developing countries show wider variation.

This highlights education as a key structural factor influencing life expectancy, as increased schooling may improve health knowledge and enable individuals to make informed health decisions. Governments should prioritise long-term investment in education systems to potentially improve life expectancy in the future. However, the boxplots do not illustrate the precise relationship between the two variables at an individual country level in a static format, meaning underlying trends are less visible.



## Visualization #4

Q4. To what extent does infant mortality contribute to differences in life expectancy between countries?



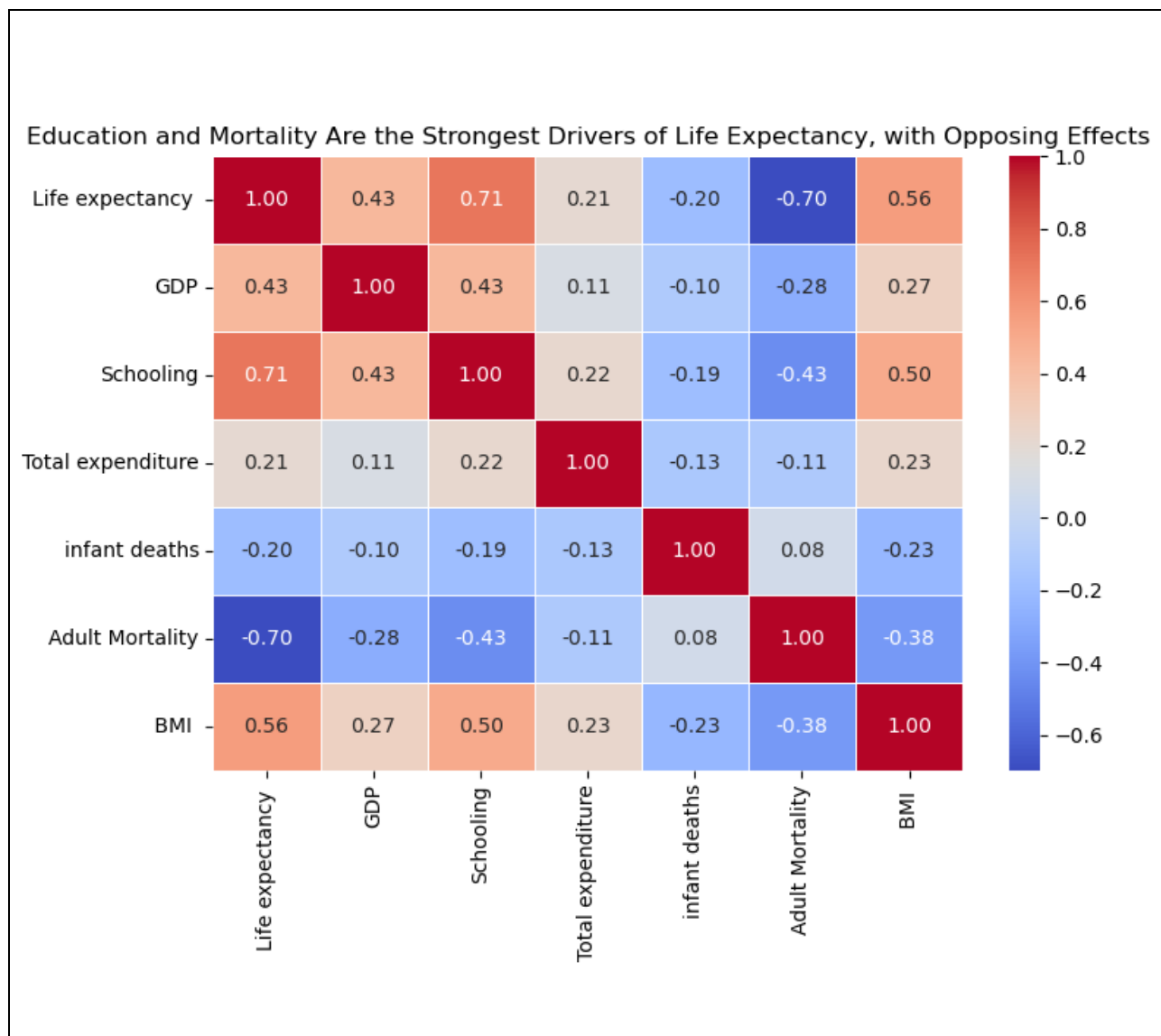
What is the main insight of the chart? (MAX 200 WORDS) Why do you think this is insightful?

The bubble chart suggests a negative relationship between infant deaths and life expectancy, with countries experiencing higher levels of infant mortality generally exhibiting lower life expectancy. This pattern is particularly evident among developing countries where there is greater variability in life expectancies and higher infant death values. This disparity can be explained by a combination of factors such as limited access to quality healthcare services and inadequate maternal care in developing regions. Developed countries are clustered at lower levels of infant deaths and consistently higher life expectancies. Including the population size further highlights that some large countries contribute significantly to global patterns of infant mortality. However, this visualisation is limited by the nature of the data, as the variable appears to represent total infant deaths rather than a standardised rate per 1000 births. Therefore, comparisons are influenced by population size, leading to extreme outliers such as India which may exaggerate the strength of the observed relationship.

The findings are insightful as they highlight the importance of early life health conditions in shaping the overall population health and influencing life expectancy. Therefore, policymakers could invest in maternal and child health programmes to improve early life healthcare.

## Visualization #5

Q5. Which factors are most strongly associated with life expectancy and how do they interact to explain global inequalities?



What is the main insight of the chart? (MAX 200 WORDS) Why do you think this is insightful?

The correlation heatmap reveals that life expectancy is influenced by a combination of interrelated socio-economic and health factors rather than a single determinant. Schooling demonstrates the strongest positive correlation with life expectancy ( $r = 0.71$ ), suggesting that education plays a critical role in improving health. This can be a result of increased health awareness and access to more resources. In contrast, adult mortality exhibits a strong negative relationship ( $r = -0.70$ ). This highlights the impact of a disease burden and healthcare effectiveness on life expectancy. While GDP is positively correlated ( $r = 0.43$ ), its correlation is weaker as compared to schooling, suggesting that economic growth alone is insufficient without effective social investment. Furthermore, the moderate correlations between variables such as GDP and schooling ( $r = 0.43$ ) indicate that these factors are connected, making it difficult to determine the cause and effect. However, the analysis is based on a correlation heatmap, meaning any non-linear relationships between variables will not be captured.

These findings suggest that policymakers and international organisations should prioritise investment in education and preventative healthcare. This is because improving early life conditions and human capital may lead to greater long-term improvements in population health than increasing healthcare spending alone.